

When Does a Transformer Learn a Reusable Semantic Relation? Competence, Depth Extrapolation, and Causal Reuse

Paper 0.6 — Semantic / Abstraction Research Line

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Abstract

We distinguish shallow success from a reusable semantic relation by training on trees of depth at most four and testing unseen identities, topologies, path lengths, branching, and distractors. Across 15 Transformer cells (depths 2–12, three seeds), parent, grandparent, k -ancestor, and root do not reach held-out competence; **isAncestor** is stronger but unstable across seeds and model depths. Its deep mean accuracy is 77.62%, and the minimum cell is 35.42%. Consequently $L_{\min}(d)$ cannot be estimated and the supported scaling class is threshold failure, not linear or logarithmic composition. Five locally competent **isAncestor** checkpoints permit bounded causal analysis: intermediate-node masking has negligible effect, root-position masking matters more, and self-attention interventions are more disruptive than feed-forward interventions, favoring a direct membership strategy over path iteration in those cells. A separate competent nested-rule task (S3: 99.93% mean, 99.22% worst cell) shows partial causal reuse: matched order-2 donors damage order-3 computation less than unrelated or cross-target donors. The result is a competence-first account: compositional depth can be measurable, but neither shallow accuracy nor geometry licenses a claim that a depth-general predicate was learned. An additional capacity diagnostic varies depth, width, head allocation, parameter allocation, and training budget. At one training budget, depth eight uniquely stabilizes **isAncestor** on easy and medium tiers but misses the hard worst-seed gate. At twice the budget, the baseline rescues easy and medium while depth eight passes all three tiers across three seeds (hard mean/worst 99.7/99.5%). Thus optimization moves the boundary and depth interacts with budget; this selected diagnostic does not isolate a depth-only scaling law or establish unrestricted depth extrapolation. A subsequent independent stress sweep finds the same measured D , d , and b frontiers for the baseline and depth-eight models, but the baseline remains competent through $N = 256$ distractors while depth eight falls just below the worst-seed gate there. Because the input exposes the candidate directly among path/edge records, these are finite contextual-membership frontiers, not evidence of iterative transitive closure. A corrected cue-free control matches candidate membership and endpoint roles across labels. On this control, $L \in \{2, 4, 8\}$ remains near chance and no model is competent even at trained depths. The compositional prerequisite therefore fails and $L_{\min}(d)$ remains undefined.

1 Question and evidential standard

Let $P(v)$ be the unique parent of a non-root node. We study

$$\begin{aligned} \text{parent}(x) &= P(x), & \text{grandparent}(x) &= P^2(x), & \text{ancestor}_k(x) &= P^k(x), \\ \text{isAncestor}(x, y) &= \mathbf{1}[\exists k \geq 1 : P^k(x) = y], & \text{root}(x) &= P^{d(x)}(x). \end{aligned}$$

A learned relation must transfer across unseen token identities and tree topologies and remain competent as total depth D , required path d , branching b , and distractor count N vary independently. Mechanism interpretation is allowed only after an 80% worst-cell gate. This ordering prevents a representation in an unsuccessful model from being redescribed as semantic computation.

For diagnostic logits z_ℓ at residual boundary ℓ , target margin is

$$m_\ell = z_\ell(y) - \max_{c \neq y} z_\ell(c).$$

For an intervention I , we report top-1 change, margin damage $m - m^I$, and Jensen–Shannon damage between output distributions. The stable boundary is the first boundary after which the target remains top-1. These quantities distinguish eventual correctness from when and how the decision is constructed.

2 Experiments

2.1 Predicate phase diagram (S1-v4)

Training contains depths 2–4 and required hops 1–3. Evaluation uses disjoint query leaves and random unseen topologies at $D \in \{2, 3, 4, 5, 6, 8, 12, 16, 24, 32\}$, hops through 12 where valid, $b \in \{1, 2, 4, 8\}$, $N \in \{0, 4, 16, 64\}$, four templates, and aligned/randomized positions. Internal relation-label vocabulary is shared so test outputs are trained symbols, while queried identities have zero overlap. The phase grid uses width 64, four heads, model depths $L \in \{2, 4, 6, 8, 12\}$, and seeds 11, 23, and 37: 15 independently trained cells and 135,360 aggregate evaluation rows.

We define

$$L_{\min}^{(\tau)}(d) = \min\{L : A(L, d, D, b, N) \geq \tau\}, \quad \tau \in \{.8, .9\}.$$

Constant, logarithmic, linear, and sublinear laws are considered only with at least four competent extrapolation points stable over seeds. Otherwise the scientifically meaningful class is threshold failure.

2.2 Competence-gated causal tests

Only five `isAncestor` cells have a deep worst cell at least .8: $(L, s) = (4, 23), (6, 11), (6, 37), (8, 11), (12, 23)$. For these local successes we decode every path token at every boundary; measure attention to path and off-path tokens; mask parent, intermediate, root, sibling, and distractor positions; zero or mean-replace SA/FF updates; and replace updates or heads with same-role donors from another tree. The other four predicates remain blocked.

2.3 Capacity and training-budget diagnostic (S1-v5)

We retain the v4 generator and compare seven allocations at one training budget: the $L4/W64/H4$ baseline; $L8/W64/H4$; $L4/W128/H4$; two and eight heads at fixed $L4/W64$; and $L12/W48/H4$ versus $L4/W96/H4$. Each uses seeds 11, 23, and 37 and is evaluated on predicate-specific easy, medium, and hard cells. We then select only the baseline and depth-eight arms for twice the training budget. Five of those six checkpoints are presently observed: all three baseline seeds and depth-eight seeds 11 and 23. We report the latter as partial evidence and do not evaluate a three-seed gate without seed 37.

2.4 Three comparison generators

The earlier controlled suite is retained because it separates kinds of generalization. S1-v2 is explicit in-context taxonomic lookup and passes (98.45% mean; 85.16% worst cell). S2 is full-vector relational parity: it reaches essentially zero training loss but only 5.66% mean and 4.69% worst-cell held-out accuracy, so all S2 mechanism claims remain blocked. S3 predicts

$$f_1 = a_1, \quad f_2 = f_1 \oplus a_3, \quad f_3 = f_2 \oplus a_6,$$

and passes unseen combinations (99.93% mean; 99.22% worst cell). Its stable decision boundary rises from 1.75 to 2.49 to 4.75 for orders one through three. We now complete its causal test by transplanting order-2 SA/FF updates into order-3 recipients using matched, unrelated same-target, and cross-target donors, plus component and head ablations.

3 Results: shallow predicate success does not extrapolate

The phase diagram overturns the tempting interpretation of the easier S1-v2 lookup task. Averaged over the deep regime, accuracies are 7.86% (parent), 7.72% (grandparent), 7.57% (root), 6.52% (k -ancestor), and 77.62% (`isAncestor`). Even `isAncestor` has a 35.42% minimum cell. The corresponding shallow means are 32.69%, 31.25%, 37.22%, 29.46%, and 81.86%. Thus four predicates fail before a depth extrapolation claim can begin; `isAncestor` displays a mixed basin of local competence rather than seed-stable predicate learning.

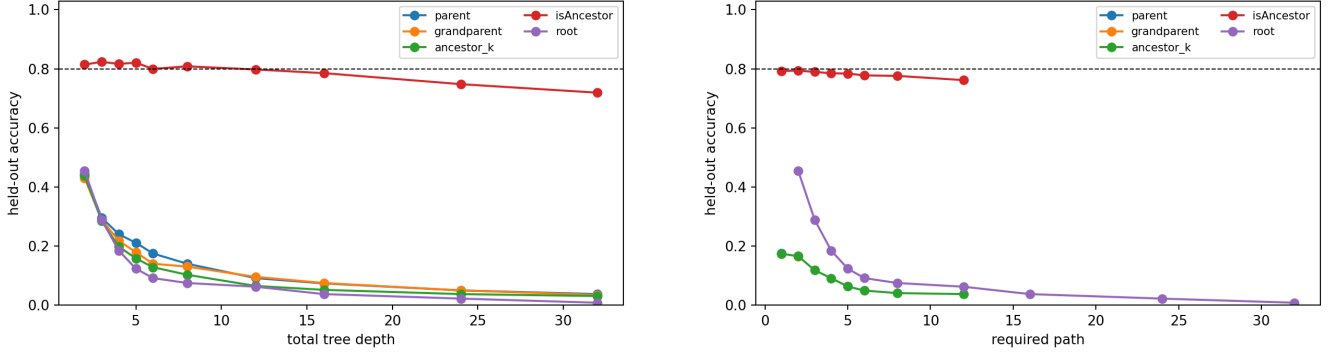


Figure 1: Held-out predicate accuracy versus total tree depth (left) and required relation path (right). The horizontal line is the competence threshold. Depth and required path are varied explicitly rather than conflated.

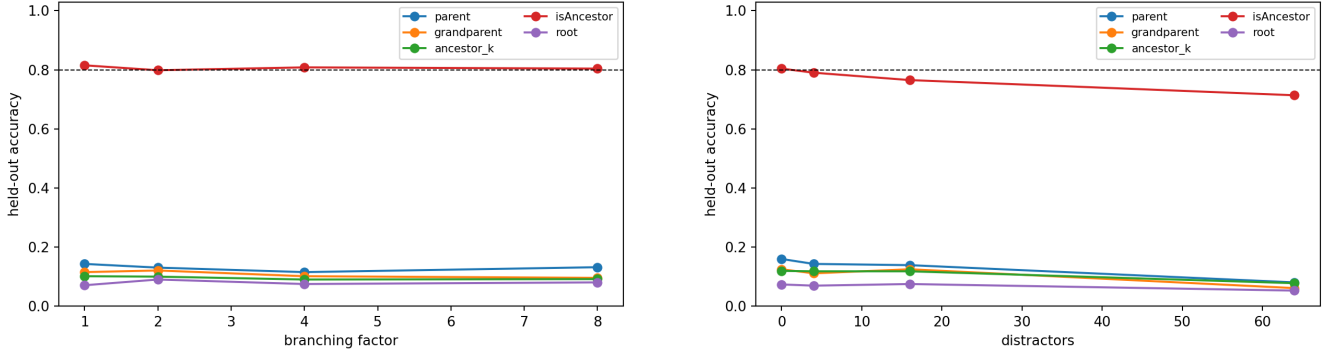


Figure 2: Independent branching and distractor controls. Neither turns the failed predicates into stable learned relations.

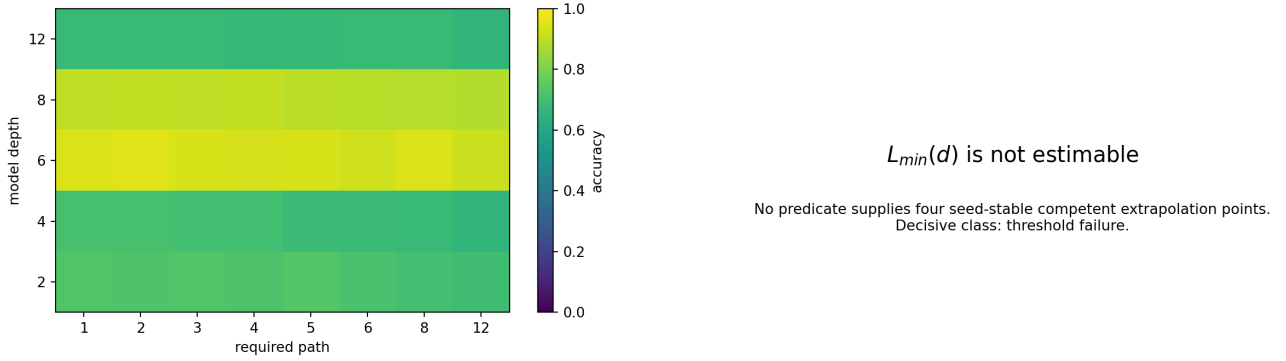


Figure 3: Left: **isAncestor** accuracy across model and path depth. Right: why no scaling curve is reported. With fewer than four seed-stable competent extrapolation points, constant, logarithmic, linear, and power fits would be numerically decorative rather than evidence.

This is the strong negative outcome anticipated by the protocol: shallow behavior reflects finite-regime interpolation. It is not evidence for one-hop-per-layer iteration, pointer doubling, or constant-depth global execution. In particular, model-depth islands cannot define $L_{\min}(d)$ when adjacent seeds fail the same cell.

4 Capacity and optimization move a narrow frontier

At one training budget, depth eight is the only tested allocation with any seed-stable competent diagnostic cells. Its **isAncestor** mean/worst-seed accuracies are 93.5/88.9%, 92.8/88.2%, and 86.9/75.9% on easy, medium, and hard tiers. It therefore passes easy and medium but not hard. Relative to baseline, depth eight raises the mean by 21.3, 21.1, and 19.3 percentage points. Doubling width changes the corresponding means by only +4.9, −0.2, +2.2 points, and its worst seed remains near chance. Eight heads improve the mean but leave a 61.1% hard-tier worst seed; two heads collapse near chance. No allocation brings parent, grandparent, k -ancestor, or root close to competence.

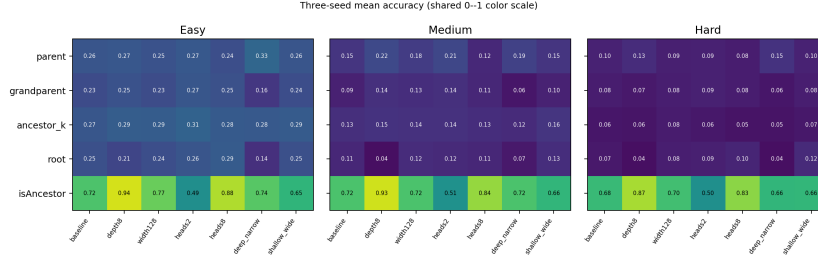


Figure 4: Observed one-budget Stage-A diagnostic. Values are three-seed means. Depth eight selectively stabilizes **isAncestor**; the four constructive predicates remain far below the 80% gate.

Parameter count alone does not explain the measured ordering. Width 128 has 579,328 parameters but fails every gate, whereas depth eight has 292,480 and passes two **isAncestor** tiers. The two-, four-, and eight-head models have identical 158,592-parameter counts, yet differ strongly in seed behavior. The deep/narrow and shallow/wide comparison is only approximate (246,048 versus 336,192 parameters), so it is an allocation control rather than an exact causal isolation of depth.

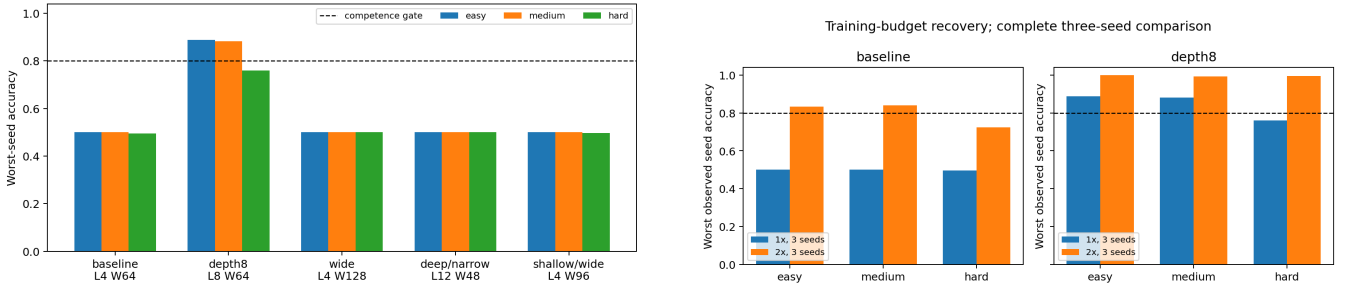


Figure 5: Left: worst-seed **isAncestor** accuracy for selected one-budget allocations. Right: complete three-seed training-budget comparison. At 2x, baseline passes easy and medium but not hard; depth eight passes all three diagnostic tiers.

Architecture	Budget/seeds	Easy	Medium	Hard	Easy–hard gap
Baseline	1x / 3	.722/.500	.718/.500	.676/.495	.046
Depth 8	1x / 3	.935/.889	.928/.882	.869/.759	.066
Baseline	2x / 3	.944/.833	.935/.840	.900/.725	.045
Depth 8	2x / 3	1.000/1.000	.998/.993	.997/.995	.003

Table 1: **isAncestor** three-seed mean/worst-seed accuracy. Gaps use tier means. The 2x rows are a selected budget control, not a complete independent frontier over D, d, b, N .

The complete baseline comparison is decisive about one issue: twice the training budget rescues easy and medium competence without additional depth. Its easy–hard mean gap remains essentially unchanged (.046 to .045), and hard worst-seed accuracy is still only .725. Thus training improves finite performance without by itself demonstrating depth-general execution. Depth eight at 2x is near-perfect for all three seeds: hard-tier mean/worst accuracy is .997/.995 and its mean easy–hard gap contracts to .003. The contrast supports a budget–allocation interaction in

this finite diagnostic: additional optimization is sufficient for the shallow model on easier tiers, whereas the deeper allocation remains seed-stable on the tested hard tier. It does not identify depth alone, because the architectures differ in parameter count and only a selected budget pair was run.

4.1 Independent measured frontiers (S1-v6)

We next vary one stressor at a time for the same baseline and depth-eight allocations, with three seeds and a strict 80% worst-seed gate. Both architectures exhaust the tested ranges $D \leq 32$, $d \leq 16$, and $b \leq 8$. Their nuisance frontiers separate: the baseline remains above threshold through $N = 256$, whereas depth eight has $N_{\max} = 64$ because its $N = 256$ worst seed is .792 (baseline .958). Mean accuracy changes from .995 in the training regime to .989 outside it for baseline, and from .998 to .980 for depth eight. Thus extra depth provides no frontier advantage here and is less robust at the most extreme nuisance setting.

Architecture	$D_{\max}$	$d_{\max}$	$b_{\max}$	$N_{\max}$	Train-regime	Extrapolation
Baseline ($L4/W64/H4$)	32	16	8	256	.995	.989
Depth 8 ($L8/W64/H4$)	32	16	8	64	.998	.980

Table 2: Largest tested value passing the three-seed worst-cell gate. These are right-censored measured frontiers: entries at the maximum tested value do not imply an unbounded capability. The four columns hold the other stressors fixed according to the preregistered sweep.

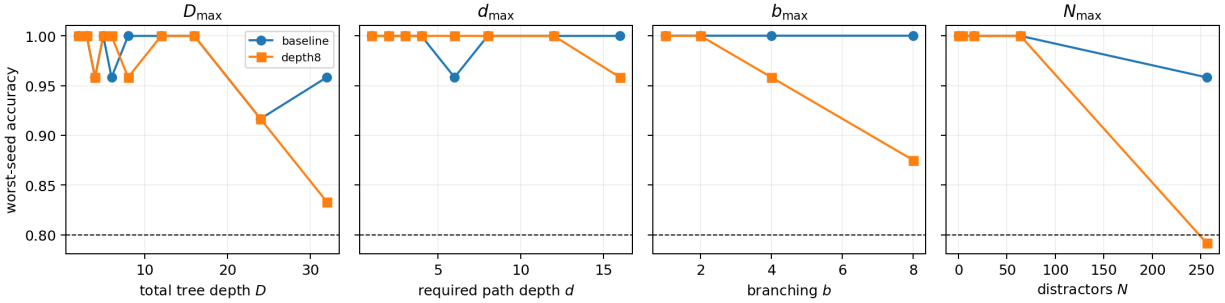


Figure 6: Worst-seed accuracy under independent total-depth, required-position, branching, and distractor sweeps. Both models remain competent across the tested D , d , and b ranges; only the depth-eight $N = 256$ cell crosses below the threshold.

The d result requires a crucial semantic qualification. In this `isAncestor` encoding, a positive candidate is itself present among the path/edge records, whereas a negative candidate is a distractor. A direct endpoint-membership comparison can therefore solve the task without composing parent links. Consequently $d_{\max} = 16$ measures robust lookup across longer presentations; it does not establish ancestor-by-ancestor execution, closure, or an estimable $L_{\min}(d)$. That interpretation also agrees with the v4 causal negative controls: intermediate-node masking is negligible and there is no stable hop progression.

4.2 Cue-free compositional control (S1-v7)

We therefore ran the narrower prerequisite experiment rather than expanding the v6 sweep. Each example contains two disjoint equal-length chains. The queried child belongs to one chain; the candidate is either its ancestor at exactly d hops or the endpoint at the same depth in the decoy chain. Thus positive and negative candidates both appear in every input, exactly once as child and once as parent. Candidate-chain order is crossed with the label, two templates shuffle all edge records, and train, validation, and test use disjoint symbol namespaces. Exact reachability supplies the label. Generator audits verify these invariants, balanced label/order cells, a .500 endpoint-membership baseline, and zero path overlap.

The $L2/W64/H4$, $L4/W64/H4$, and $L8/W64/H4$ models use seeds 11, 23, and 37. Training draws $d \leq 3$; held-out evaluation uses $d \in \{1, 2, 3, 4, 6, 8, 12, 16\}$ with four templates, aligned/randomized placement, and three topology seeds. Competence requires at least .8 for every seed and every template-position-topology stratum. Aggregate test accuracy is approximately chance throughout (Table 3); critically, no architecture passes even $d = 1$. The

preregistered prerequisite for extrapolation is therefore absent: all frontiers are zero, fewer than four competent extrapolation depths exist, and $L_{\min}(d)$ is not eligible for estimation.

Architecture	Mean accuracy	Seed-depth range	Competent depths
$L2/W64/H4$	.496	.380–.562	none
$L4/W64/H4$	.502	.370–.661	none
$L8/W64/H4$	.498	.424–.581	none

Table 3: Cue-free v7 test accuracy across three model seeds and eight required depths. The range is over seed–depth aggregate cells; the competence decision is stricter and checks every template, position, and topology stratum. Chance is .5.

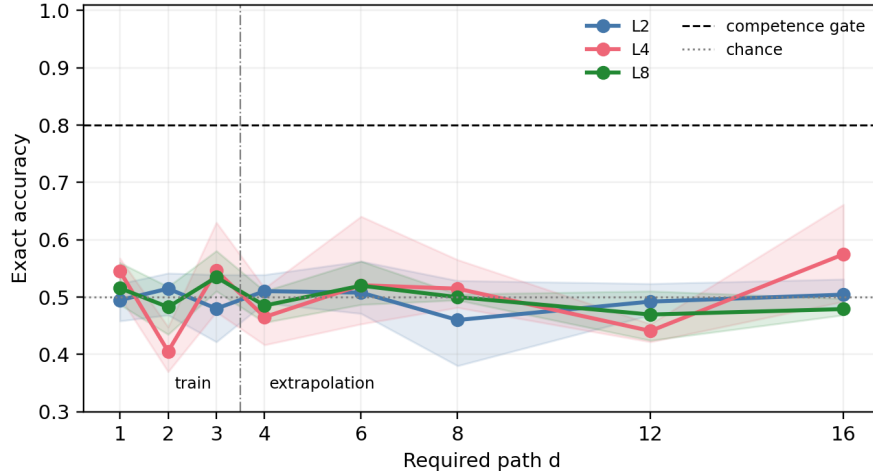


Figure 7: Cue-free v7 aggregate accuracy by required path. Curves are three-seed means and bands span seed minima to maxima. Neither trained depths (solid markers, $d \leq 3$) nor extrapolation depths approach the .8 gate.

This is a clean acquisition failure in the tested architecture and optimization regime, not proof that Transformers cannot compute reachability. It does show that the positive v6 path-length result does not survive removal of its endpoint-membership cue. Because the easiest trained depth fails, deeper or broader scaling is stopped rather than interpreted as a compositional frontier.

5 Local S1 mechanism: direct membership, not path iteration

The five competent `isAncestor` cells are useful local case studies, not a rescue of the central gate. Intermediate-candidate masking produces mean JS damage below 10^{-4} , whereas masking the root position produces 0.0018. Boundary decoding and attention do not exhibit a stable 1, 2, 4, 8 hop progression. Together these results favor early/direct endpoint comparison or global membership lookup over a necessary sequential traversal of intermediate ancestors.

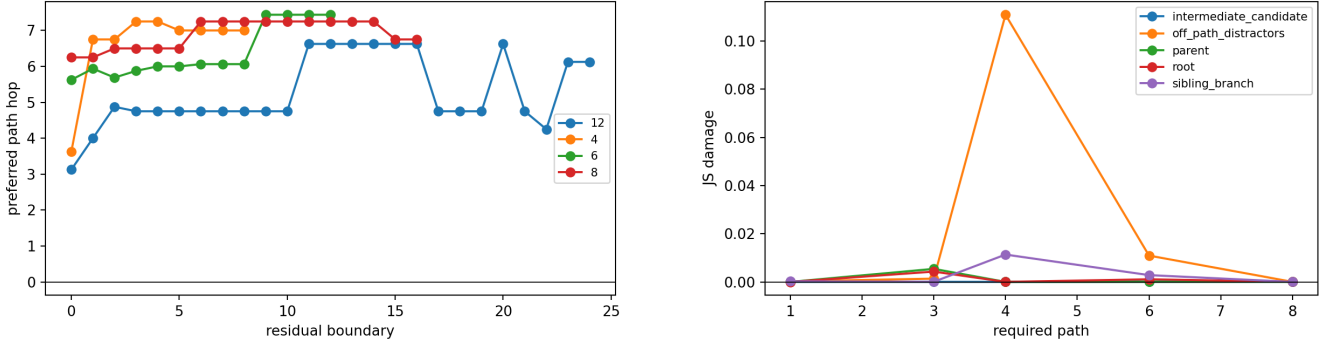


Figure 8: Locally competent `isAncestor` cells only. Preferred decoded path position (left) lacks a seed-stable iterative trajectory; intermediate masking is nearly harmless compared with endpoint masking (right).

Same-role cross-tree update replacement causes mean target-margin damage 0.2918: relational role is not freely substitutable across contexts. SA interventions have mean JS damage 0.0619 versus 0.0054 for FF, an approximately eleven-fold contrast. This localizes sensitivity to contextual transport, but does not prove that any individual attention map is an explanation.

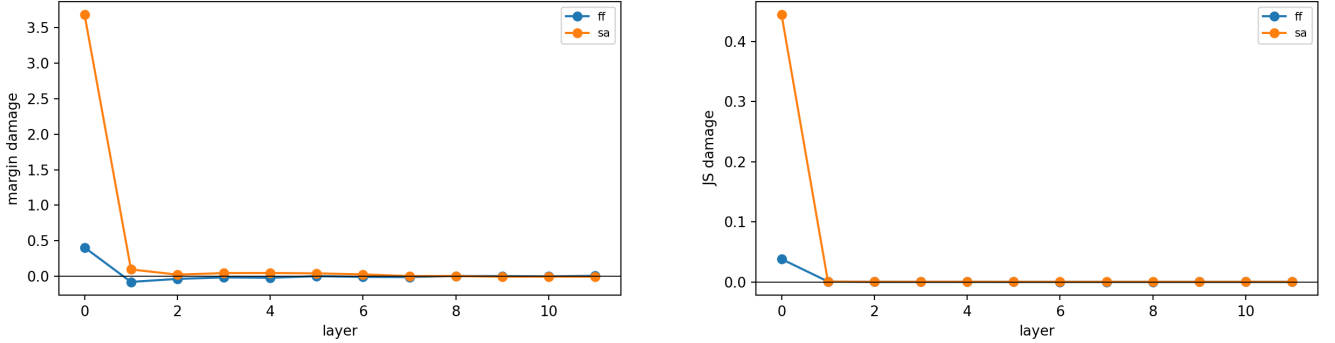


Figure 9: Cross-tree role replacement (left) and component intervention (right), restricted to competent local cells. SA is substantially more causally sensitive than FF.

6 S3 causal reuse of nested computation

For competent S3, replacing order-3 updates by matched order-2 donors causes mean margin damage 3.6449. An unrelated donor with the same semantic target causes 4.3543, and a cross-target donor causes 6.2256; top-1 flip rates are 12.50%, 17.19%, and 23.96%, respectively. The ordered control matters: target agreement alone does not explain the advantage of matched donors. Order-2 computation is therefore partially reusable inside order-3, although nonzero matched damage shows that the representation remains stage- and context-sensitive.

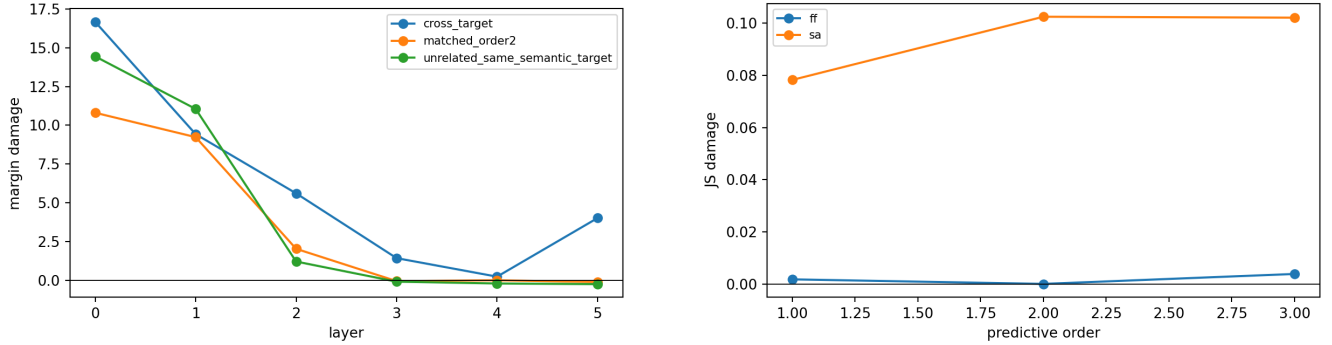


Figure 10: S3 causal completion. Role-matched order-2 donors are less damaging than cross-target controls (left); SA/FF sensitivity changes with predictive order (right).

The earlier stable-boundary sequence 1.75, 2.49, 4.75 and the donor ordering jointly support a deliberately local theory: more target-relevant composition consumes later computation, partly by reusing lower-order structure. They do not establish a universal hierarchy of semantic abstraction.

7 Scorecard and interpretation

No S1-v4 predicate passes most dimensions of the operational scorecard. Identity separation is guaranteed by construction, but parent, grandparent, k -ancestor, and root fail held-out topology competence. After targeted training, **isAncestor** becomes seed-stable throughout most of the v6 measured membership frontier, but this does not repair mechanism consistency or demonstrate compositional depth generalization. Its classification is *finite*, *direct-membership behavior*. In v7, matching candidate exposure removes that shortcut and all three tested depths fail the acquisition prerequisite; the constructive predicates remain global threshold failures. S2 remains a memorization negative control. S3 supplies positive evidence for bounded compositional reuse.

The combined lesson is pedagogically simple. First ask whether the rule works beyond its training support. Then ask how the output is constructed. Finally require interventions that distinguish relational role from token identity and target agreement. Reversing this order lets attractive trajectories or clusters assign semantics to a failed computation.

8 Limitations and reproducibility

The experiments use synthetic symbolic relations, and even the independent sweeps are finite, right-censored probes. Checkpoints are locally reproducible but excluded from version control; all reported numeric claims are backed by tracked raw CSVs. The v4 configuration is `configs/paper06/predicate_v4.json`; the capacity design is `configs/paper06/capacity_v5.json`; the v6 stress design is `configs/paper06/stress_v6.json`; and the cue-free design is `configs/paper06/compositional_frontier_v7.json`. Generator and model support are in `src/cl/semantic/predicates.py`, `src/cl/semantic/predicate_stress.py`, `src/cl/semantic/predicate_composition.py`, and `src/cl/common/model_adapter.py`. The v7 runner is `paper06_compositional_frontier.py`. Its nine official models completed on CPU in 26.7 aggregate model-minutes; the manifest records device, configuration and generator hashes, all gates, and exact sample counts. The v6 runner and analyzer are `paper06_stress_frontier.py` and `paper06_stress_analyze.py`. Its six official cells were completed on CPU after an incomplete MPS attempt repeatedly stalled during optimization; the incomplete MPS run is excluded from every reported statistic. Tracked manifests, raw evaluations, stable-cell summaries, and figures identify the clean CPU runs and their exact sample counts.

9 Conclusion

A model can solve explicit taxonomic lookup without learning a predicate that extrapolates in depth. In the harder shallow-train/deep-test instrument, four constructive predicates remain incompetent across the tested capacity allocations. **isAncestor** has a movable finite membership boundary: optimization rescues diagnostic tiers, and both selected architectures exhaust the tested D , d , and b ranges, yet extra depth gives no frontier advantage and loses the strict $N = 256$ gate. Once candidate membership and endpoint roles are matched, all v7 architectures remain near

chance even at trained depths. This blocks, rather than estimates, a compositional scaling law. Local `isAncestor` causality likewise favors direct contextual membership over ancestor-by-ancestor iteration, whereas competent S3 provides controlled evidence that lower-order computation is partially reused within a higher-order rule. Capacity, optimization, extrapolation, and causal role invariance are separate requirements; only their conjunction supports a reusable semantic relation.

A Legacy six-tree pilot

The original six-tree natural/arbitrary pilot is preserved as a negative control. Across four trained runs its final behavioral accuracy was 1.4–15.3%, yet post-block normalized separation was 0.110, hierarchy RSA 0.194, and parent-neighbor recovery 0.477 versus 0.254 under permutation. Cross-template cosine reached 0.960. These above-null geometries in an incompetent model motivate the present competence gate. Its 4,320 representation rows, 2,880 component rows, 4,320 entropy rows, 5,760 replacement rows, 2,304 competence rows, 108 aggregates, and 13 figures remain tracked in the legacy artifact tree.

B Pretrained and continual-learning boundary

The pretrained audit remains external-validity material rather than evidence for the controlled claim: training exposure, ontology, and frequencies are unknown. WordNet-style siblings, paraphrase invariance, residual-matched donors, and JVP tests may probe graph recovery but cannot reproduce generator-level counterfactual control. Continual-learning implications are likewise downstream. Failed S2 or unstable S1-v4 features should not be promoted to persistent invariants; competent S3 features are candidates only within the demonstrated generator and causal scope.